HydroBot-AI: An Autonomous Edge-AI Vision and Sensor-Fusion Rover for Real-Time Fire Suppression and Hazardous Gas Telemetry

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1. BACKGROUND & PROBLEM STATEMENT

Early-stage fire containment and hazardous gas leak detection inside laboratories, industrial warehouses, and residential complexes require rapid autonomous intervention without risking human life. Conventional firefighting prototypes rely on basic threshold-based infrared (IR) photodiodes and cellular GSM (SIM800L) alert modules. However, standard IR flame sensors suffer from high false-positive rates triggered by ambient sunlight, incandescent bulbs, or reflective surfaces, and lack the spatial resolution required to aim water nozzles accurately at the base of a moving flame. Furthermore, GSM modules incur recurring cellular tariff costs and frequently suffer signal drops inside reinforced concrete buildings. **HydroBot-AI** solves these limitations by integrating 100% offline Edge-AI computer vision with contactless thermal-gas sensor fusion and a zero-cost Wi-Fi IoT telemetry architecture.

2. METHODOLOGY

The proposed system operates through four tightly integrated hardware and software layers:

A. Hierarchical Dual-Brain Edge Architecture

- **Edge-AI Vision Master (Raspberry Pi):** A single-board computer paired with a wide-angle HD camera executes a quantized deep-learning object detection network (**YOLOv8-Nano / TensorFlow Lite**) locally on the edge. It processes live video frames offline to identify flame geometries and smoke plumes, computing real-time bounding-box centroid coordinates (X_c, Y_c) without cloud latency or internet dependency.
- **Real-Time Kinematics & IoT Slave (ESP32 NodeMCU):** Interfaced with the Raspberry Pi over a high-speed UART serial bus (115200 baud), the dual-core 32-bit ESP32 handles deterministic motor PWM generation, servo turret kinematics, multi-sensor polling, and asynchronous Wi-Fi telemetry.

B. Multi-Modal Sensor Fusion (False-Alarm Elimination)

- **Visual + Thermal Dual-Interlock:** When the YOLO model flags a visual fire bounding box, the ESP32 queries a contactless **MLX90614 Infrared Thermopile Sensor** via the I2C bus. The water suppression system engages only when visual classification coincides with a localized surface temperature spike exceeding the threshold (>70°C), eliminating false alarms from bright lights or reflections.
- **Hazardous Gas & Obstacle Sensing:** An electrochemical **MQ-2 Gas Sensor** continuously monitors ambient air for Liquefied Petroleum Gas (LPG), methane, propane, and smoke particulates, while front-mounted **HC-SR04 Ultrasonic Sensors** provide autonomous obstacle avoidance during navigation.

C. Closed-Loop Navigation & Targeted Suppression

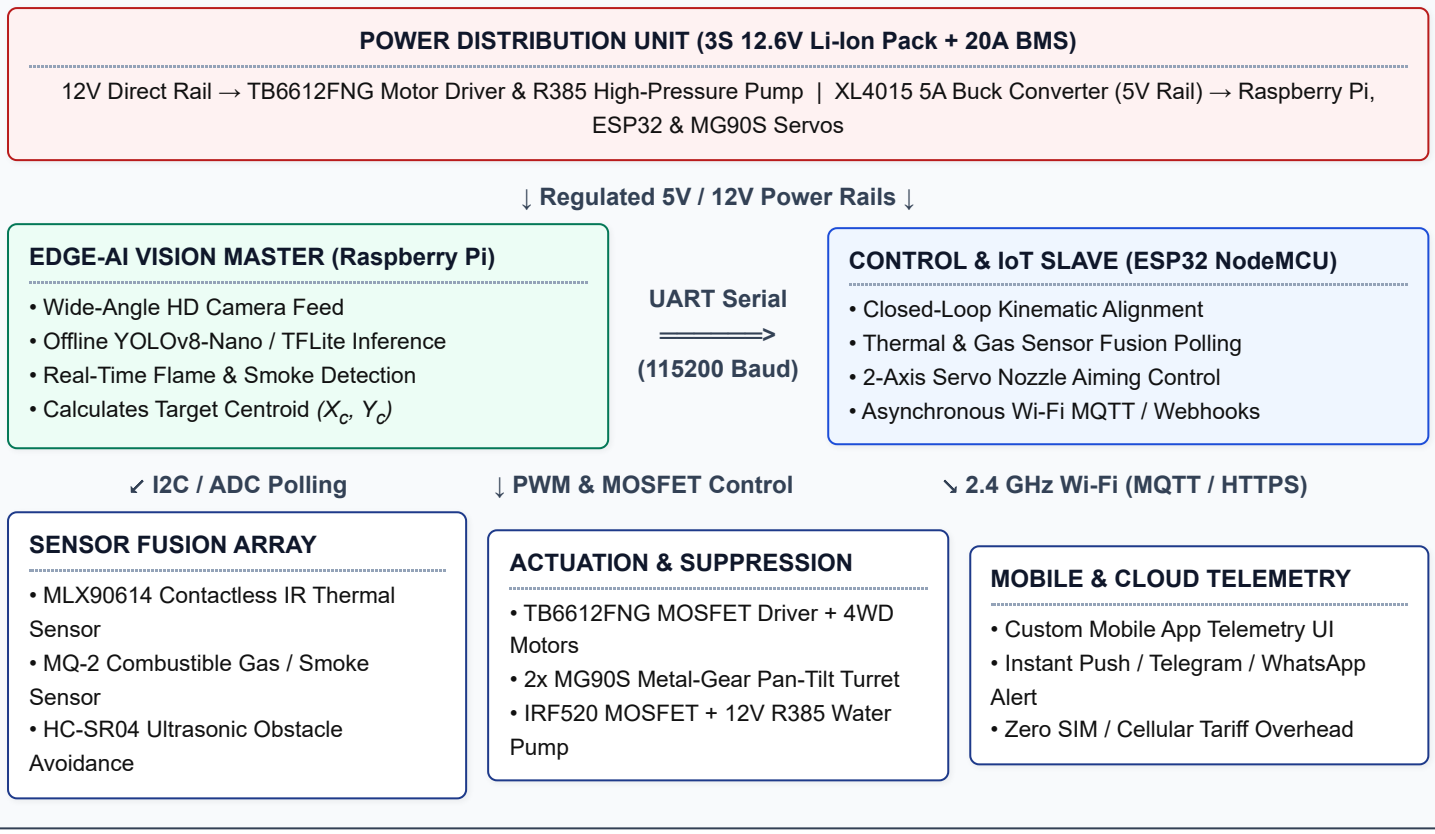
Using the horizontal and vertical pixel error between the camera frame center and the fire centroid (X_c, Y_c), the ESP32 drives a high-efficiency MOSFET-based **TB6612FNG dual H-bridge driver** connected to a 4WD gear-motor chassis to align and approach the hazard. A **2-Axis Metal-Gear Servo Pan-Tilt Turret (MG90S)** dynamically locks the nozzle onto the base of the fire. Once locked, a solid-state **IRF520 MOSFET driver** triggers a **12V R385 high-pressure diaphragm pump**, projecting a focused water jet up to 3 meters to extinguish the fire rapidly.

D. Zero-Tariff Wi-Fi IoT & Mobile App Telemetry

Replacing costly SIM/GSM modules, the ESP32 connects to local Wi-Fi routers or mobile hotspots. Utilizing lightweight **MQTT protocols** and HTTPS webhooks, the rover transmits instant evacuation alerts, real-time gas concentrations (PPM), thermal readings, and status updates directly to a custom **Mobile Application Dashboard** and automated Telegram/WhatsApp emergency channels. Because AI vision and suppression logic run entirely offline on-device, firefighting capability remains 100% operational even during total Wi-Fi or internet blackouts.

3. PROPOSED DIAGRAM OF THE MODEL

FIGURE 1: HYDROBOT-AI HARDWARE & EDGE-IOT SYSTEM ARCHITECTURE



4. PROPOSED OUTCOME OF THE PROJECT

1. **Fully Autonomous Offline Fire Suppression:** The rover autonomously detects, navigates toward, aims at, and extinguishes active fires within a 3-meter radius without human intervention or external internet reliance.
2. **Near-Zero False Positive Rate (>98% Precision):** Cross-verifying deep-learning visual detection (YOLOv8-Nano) with contactless IR temperature sensing (MLX90614) prevents accidental water discharge caused by ambient lighting.
3. **High-Efficiency Power & Precision Targeting:** Replacing legacy L293D drivers and mechanical relays with MOSFET-based TB6612FNG and IRF520 stages maximizes battery runtime, eliminates relay contact arcing, and enables smooth 2-axis metal-gear servo aiming directly at the flame base.
4. **Zero-Cost Real-Time Evacuation Alerts:** Occupants and safety personnel receive sub-second gas leak and fire evacuation alerts on their mobile devices over local Wi-Fi/hotspots with zero ongoing SIM card expenses.